

C1 - M1 - S1 (Basics of Probability)

Q1. What is the probability of India winning the match **AND** Sachin scoring a century?
(Answer expected in **X.XX** format)

Refer to the table below:

	India Win	India Lose	Total
Sachin Scores a Century	10	2	12
Sachin Doesn't Score a Century	50	38	88
Total	60	40	100

Answer: 0.1

Q2. What is the probability of India losing the match **AND** Sachin not scoring a century?
(Answer expected in **X.XX** format)

Refer to the table below:

	India Win	India Lose	Total
Sachin Scores a Century	10	2	12
Sachin Doesn't Score a Century	50	38	88
Total	60	40	100

Answer: 0.38

Q3. Let us redefine the random variable X as:

$$X = 1.5 \times (\text{Number of red balls the player gets after playing the game})$$

For example, if the player gets 1 red ball out of the 4 total balls, then

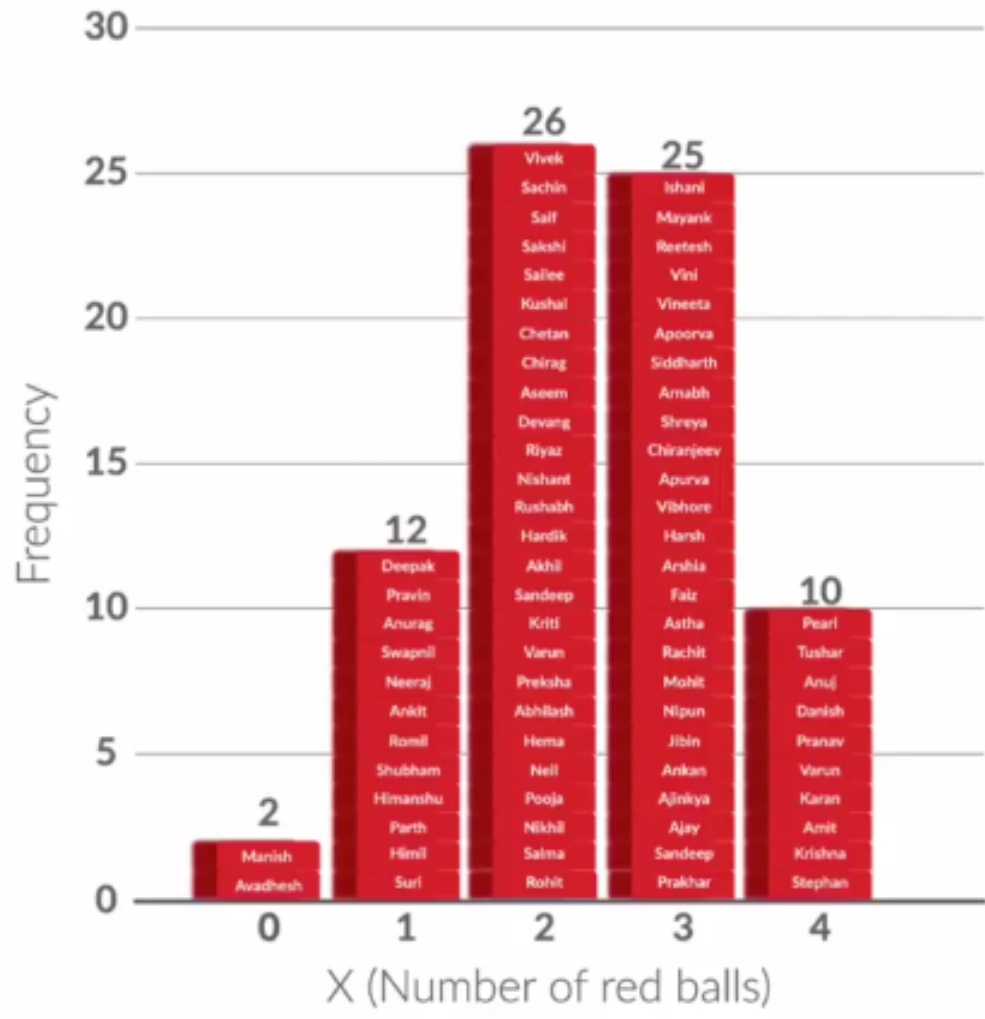
$$X = 1 \times 1.5 = 1.5$$

Out of the 16 possible outcomes of this game, how many outcomes will result in $X = 3$?

Options:

- 6 $\rightarrow$ CORRECT
- 4
- 1
- 0

Q4. Based on the histogram shown below, can you determine the probability for $X = 2$?



(Give your answer as a number rounded to 2 digits after the decimal point. For example, instead of 90%, type 0.90.)

Answer: 0.31

Q5. Which of the following probability distributions is/are NOT valid complete probability distribution(s)?

A

X	Probability
0	0.1
2	0.2
3	0.3
4	0.4

B

X	Probability
0	0.15
0.5	0.15
1	0.2
1.5	0.25
2	0.25

C

X	Probability
0	-0.05
1	0.20
2	0.25
3	0.35
4	0.40

Options:

- Only Table A is an invalid distribution.
- Only Table B is an invalid distribution.
- Only Table C is an invalid distribution. → CORRECT

Q6. Calculate the expected value of X (where X is the money won by the player after playing the game once).

X (Money won after playing once)	Probability
+150	0.133
-10	0.867

You can use this probability distribution to find the answer.

Answer: 11.28

Q7. So, the expected value of a player's winnings is positive, but we want to make it negative. Which of the following changes to the game would make the expected value negative?

- Increase the penalty to Rs. 25 $\rightarrow$ CORRECT
- Decrease the penalty to Rs. 90

Q8. Suppose we change the game's rules to the following:

Outcome	Prize
4 red balls	+150
4 blue balls	-150
Any other outcome	-10

What will be the expected value for X (the amount of money won by a player after playing the game once)?

Options:

- -2.5
- -5
- +2.5
- +7.5 $\rightarrow$ CORRECT

Rajya Laxmi Bank has given student loans to 10,000 people. The government wants to ensure that the bank does not give away very risky loans and therefore wants to estimate the **Expected Loss** of the bank's student loan portfolio.

The expected loss is defined as the expected value of the money lost by the bank due to customers defaulting on their loans (i.e., not paying their EMIs).

The data used by the bank to calculate the expected loss is shown below:

Customer No.	Exposure at Default (Rs lakh)	Recovery (%)	Probability of Default
00001	11.50	20%	0.007

Here:

- **Exposure at Default (EAD)** is the total money owed by the customer in case of default.
- **Recovery (R)** is the percentage of the exposed amount that the bank is able to recover.
- **Probability of Default (PD)** is the probability that the customer will default on the loan.

Q9. To calculate the expected loss, the bank has created the following table. As you already know, the probability that customer 1 being a defaulter is 0.007 and that the customer is not a defaulter is 0.993.

In the event of default, what would be the value of X , i.e., the money lost by the bank?

Defaulter / Non-defaulter	Probability	X (Money Lost by Bank)
Defaulter	0.007	—
Non-defaulter	0.993	0

Options:

- Rs 2.3 lakh
- Rs 9.2 lakh → CORRECT
- Rs 11.5 lakh
- Rs 57.5 lakh

Q10. Now, the bank has the following data:

Defaulter/Non-defaulter	Probability	X (Money Lost by Bank)
Defaulter	0.007	Rs. 9.2 lakh
Non-defaulter	0.993	0

What would be the expected loss (expected value of the money lost by the bank)?

- Rs. 6440 $\rightarrow$ CORRECT
- Rs. 9.2 lakh
- Rs. 9200
- Rs. 6.44 lakh

Q11. So, in general, what would the expected loss be? [Note, Probability of Default: PD, Exposure at Default: EAD, Recovery rate: R]

- Probability of Default $\times$ Exposure at Default $\times$ Recovery Rate
- Probability of Default $\times$ Exposure at Default $\times$ (1-Recovery Rate) $\rightarrow$ CORRECT
- (1-Probability of Default) $\times$ Exposure at Default $\times$ Recovery Rate

Q12. Now, the bank wants to find the total expected loss. This is done by calculating the expected losses for each of the 10,000 customers and then adding them. Data for all bank customers are given in the file ‘Student Loan.csv’. What would be the value of the total expected loss?

- Rs. 1.25 crores
- Rs. 4.93 crores $\rightarrow$ CORRECT
- Rs. 6.18 crores
- Rs. 3.09 crores

Courses	DS	ML	DL	BD	AI	Total
Male	80	60	40	50	30	260
Female	70	40	50	70	10	240
Total	150	100	90	120	40	500

Q13. Given this contingency table, what is the probability that a randomly selected person joined Data Science?

- $\frac{80}{260}$
- $\frac{70}{240}$
- $\frac{150}{260}$
- $\frac{150}{500} \longrightarrow \text{CORRECT}$

Q14. Given this contingency table, what is the probability that a randomly selected female joins DS? In other words, what is the probability of a person joining DS, GIVEN that she is female?

- $\frac{40}{90}$
- $\frac{150}{240}$
- $\frac{240}{500}$
- $\frac{70}{240} \longrightarrow \text{CORRECT}$

Q15. Consider a set containing all DL students OR all male students. What is the Probability that a randomly selected person will belong to this set?

Hint: Use the formula: $p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B)$

- $\frac{40}{90}$
- $\frac{50}{90}$
- $\frac{310}{500} \longrightarrow \text{CORRECT}$
- $\frac{260}{500}$

Q16. What is the probability that a student being a female AND an AI student?

Hint: Use the formula: $P(A \text{ AND } B) = P(A|B) \times P(B)$

- $\frac{10}{500} \longrightarrow \text{CORRECT}$
- $\frac{30}{500}$
- $\frac{40}{500}$
- $\frac{10}{30}$

Q17. What is the probability that a Deep Learning (DL) student being a male?

- $\frac{40}{500}$
- $\frac{90}{500}$
- $\frac{40}{90} \longrightarrow \text{CORRECT}$
- $\frac{130}{500}$

(Roulette Wheel)

Roulette wheel is a game found in many casinos. The European roulette contains the numbers 0 to 36 written in an irregular sequence. The players can bet on any number from 0 to 36. For example, let us say that Kriti bets £ 100 on the number 5. Now, a ball would drop onto the wheel, which is then given a spin. If the ball lands in the pocket marked 5, Kriti would win $(£ 100) \times 36 = £ 3,600$, resulting in net winnings of $£ 3,600 - £100 = £3,500$. However, if the ball lands on any other pocket, Kriti would not win anything, resulting in net winnings of $£ 0 - £ 100 = -£ 100$.

The American roulette wheel, in addition to the numbers 0 to 36, contains a number '00'. The player can bet on any number from 1 to 36.

Q18. European Roulette: What is the probability that Kriti wins the game if she bets on the number 5?

- $\frac{1}{36}$
- $\frac{1}{37} \rightarrow \text{CORRECT}$
- $\frac{5}{36}$
- $\frac{5}{37}$

Q19. European Roulette: Let us say that we define the random variable X as Kriti's net winnings if she plays this game and bets Rs. 100 on the number 5. What is the expected value of X?

- Rs. 97.30
- -Rs. 2.70 $\rightarrow \text{CORRECT}$
- Rs. 98.15
- -Rs. 1.85

Q20. American Roulette: Let us again define the random variable X as Kriti's net winnings if she plays this game and bets Rs. 100 on the number 5. What is the expected value of X this time?

- Rs. 97.30
- -Rs. 2.70
- Rs. 94.74
- -Rs. 5.26 $\longrightarrow$ CORRECT

C1 - M1 - S2 (Discrete Probability)

The bag contains 2 blue and 3 red balls. One trial = taking out a ball, noting its color, and putting it back in the bag.

Q21. What is the probability that you would get this combination of balls after 4 trials?

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i.e., the combination Blue-Red-Red-Red.

- $0.4 \times 0.4 \times 0.4 \times 0.4$
- $0.6 \times 0.6 \times 0.6 \times 0.6$
- $0.4 \times 0.6 \times 0.6 \times 0.6 \longrightarrow$ CORRECT
- $0.4 \times 0.4 \times 0.6 \times 0.6$

Q22. What is the probability that you will get 3 red balls after 4 trials?

- $0.4 \times 0.6 \times 0.6 \times 0.6$
- $2 \times (0.4 \times 0.6 \times 0.6 \times 0.6)$
- $3 \times (0.4 \times 0.6 \times 0.6 \times 0.6)$
- $4 \times (0.4 \times 0.6 \times 0.6 \times 0.6) \longrightarrow$ CORRECT

Q23. What is the probability that you would get the combination of balls shown below after 4 trials?



Note that the bag contains 4 red balls and 1 blue ball.

- $0.4 \times 0.6 \times 0.6 \times 0.6$
- $0.2 \times 0.2 \times 0.2 \times 0.2$
- $0.8 \times 0.8 \times 0.8 \times 0.8$
- $0.2 \times 0.8 \times 0.8 \times 0.8 \longrightarrow \text{CORRECT}$

Q24. What is the probability that you would get exactly 3 red balls after 4 trials?

- $0.2 \times 0.8 \times 0.8 \times 0.8$
- $2 \times (0.2 \times 0.8 \times 0.8 \times 0.8)$
- $3 \times (0.2 \times 0.8 \times 0.8 \times 0.8)$
- $4 \times (0.2 \times 0.8 \times 0.8 \times 0.8) \longrightarrow \text{CORRECT}$

Q25. What is the probability that you would get 4 red balls after 4 trials, if the probability of getting a red ball in one trial is equal to p ?

- $4p$
- $p^4 \longrightarrow \text{CORRECT}$
- $4p^3$
- $p^3(1 - p)$

Q26. A student attempts a test with 10 multiple choice questions, where each question has 4 possible options. She does not know the correct answer to any of the questions and just randomly selects one of the 4 options for each of them. The probability that she would still get 3 questions correct is:

- ${}^{10}C_3 \left(\frac{1}{4}\right)^{10}$
- ${}^{10}C_3 \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^7$
- ${}^{10}C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^7 \longrightarrow \text{CORRECT}$
- Binomial Probability cannot be used here

Q27. Now, the test is modified by adding 5 true and false questions to the set of 10 MCQs. The student does not know the answer to the new questions either and randomly selects true or false for each of them, too. Now, according to the binomial distribution, the probability that she would still get 5 of the 15 questions (5 true/false Qs and 10 MCQs with 4 options) correct is:

- ${}^{15}C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^{10}$
- ${}^{15}C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{10}$
- ${}^{15}C_5 \left(\frac{1}{4}\right)^3 \left(\frac{1}{2}\right)^2 \left(\frac{3}{4}\right)^{10}$
- Binomial Probability cannot be used here $\longrightarrow \text{CORRECT}$

Q28. Recall the upGrad game with the red and blue balls. For this game, can you find the probability of getting 3 or fewer red balls, i.e. $P(X \leq 3)$?

x (Number of Red Balls)	P(x)
0	0.0256
1	0.1536
2	0.3456
3	0.3456
4	0.1296

- 0.8704 $\rightarrow$ CORRECT
- 0.1296
- 0.7408

Q29. Let us define X as the number of wickets Ishant Sharma would take in the next T20 match he plays. In addition to this, the following is an incomplete table for the cumulative probability :

x (Number of wickets)	F(x)
0	0.35
1	0.55
2	0.75

What is the probability that he would take more than 2 wickets in the next T20 match he plays?

- 0.75
- 0.25 $\rightarrow$ CORRECT
- 0.50
- 1

Q30. Generally, when a user visits a website, they are offered a lead form. The following is the entire data for an upcoming e-commerce firm regarding the website visit time of its users:

Time of the Day	Cumulative Leads
1100 hours	3
1200 hours	15
1300 hours	26
1400 hours	51
1500 hours	61
1600 hours	78
1700 hours	95
1800 hours	105
1900 hours	115
2000 hours	124

What is the approximate probability that the lead had visited the platform between 1500 hours and 1600 hours?

- 0.14 → CORRECT
- 0.17
- 0.20
- 0.40

Q31. Based on the provided data, what is the approximate probability that the lead had visited the platform by 1600 hours?

- 0.14
- 0.49
- 0.63 → CORRECT
- 0.77

(FSSAI Case)

Let us say you get a job as a senior analyst at a food regulatory body such as the FSSAI. Recently, Kwick Foods filed a complaint against Sunshine, its competitor, stating that Sunshine's ready-to-cook pasta contains excess amounts of lead. Suppose that you take 10 random pasta packets from the market and have them tested for lead amount. There is a 5% probability that any pasta packet you select is going to be defective.

Q32. What is the probability that 2 of the 10 packets tested will be defective?

- 1.66%
- 7.49% → CORRECT
- 10%
- 15.92%

Q33. What is the probability that, after testing these 10 packets, not more than 2 packets would turn out to be defective?

- 98.84% → CORRECT
- 75.42%
- 82.68%
- 90.65%

Q34. Let us define X as the number of packets found to be defective after the 10 packets have been tested. What will be the expected value of X ?

- 1
- 0.5 → CORRECT
- 55
- 2

(Cancer Treatment)

Suppose a new cancer treatment has been discovered, claiming to increase the one year survival rate for pancreatic cancer to 40%. The hospital has a total of 10 patients suffering from pancreatic cancer.

Q35. What is the probability that exactly 4 of these patients would survive the first year after receiving this treatment?

- 2.2%
- 12.6%
- 25.1% → CORRECT
- 21.6%

Q36. What is the probability that the number of patients who survive the first year after receiving the treatment would not be more than 2?

- 14.9%
- 16.7% → CORRECT
- 22.1%
- 15.6%

C1 - M1 - S3 (Continuous Probability Distributions)

Q37. We have a fictional company with 3,000 employees in a local office. Let us say you went to that office and asked those 3,000 employees to play our upGrad red ball game. Based on the data on these 3,000 people, if you create the probability distribution for X , then the values of $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, $P(X = 3)$ and $P(X = 4)$ would all be:

[X = number of red balls drawn by a player after playing the game once]

- Zero
- Non-Zero $\rightarrow$ CORRECT

Q38. Suppose that you work at a sports analysis company and you want to analyze the effect a bowler's height has on his/her performance. Based on the data, you conclude that the cumulative probability, $F(175.3 \text{ cm}) = 0.3$. In this case, which of the following statements is correct?

- $P(X < 175.3 \text{ cm}) = 0.3$
- $P(X \leq 175.3 \text{ cm}) = 0.3$
- Only Statement I is correct
- Only Statement II is correct
- Statements I and II are correct $\rightarrow$ CORRECT

Q39. Suppose that you have to find the cumulative probability for a random variable X , which is normally distributed. You do not know the values of μ and σ . Can you find the cumulative probability at the point $X = \mu + \sigma$, i.e., the probability of the variable being less than ' $\mu + \sigma$ '?

- Yes $\rightarrow$ CORRECT
- No

Q40. What is the probability that a normally distributed random variable lies within 1.65 standard deviations of the mean? (Use the Z-Table)

- 95%
- 90% → CORRECT
- 85%
- 80%

(Pharma Company)

Let us say that you work as an analyst at a pharma company which manufactures an antipyretic drug (tablet form) with paracetamol as the active ingredient. The amount of paracetamol specified by the drug regulatory authorities is 500 mg with a permissible error of 10%. Anything below 450 mg would be a quality issue, while anything above 550 mg would be a serious regulatory issue.

Q41. The regulatory authority selects a random tablet from Batch Z2. Based on previous knowledge, you know that Batch Z2 has a mean paracetamol level of 510 mg and its standard deviation is 20 mg. What is the probability that the tablet selected by the authority has a paracetamol level below 550 mg?

- 48%
- 95%
- 98% → CORRECT
- 93%

Q42. Now, the company's Quality Control(QC) department arrives and selects a tablet at random from Batch Z2. What is the probability that the tablet selected by QC has a paracetamol level above 450 mg?

- 99.87% → CORRECT
- 99.47%
- 49.87%
- 99.61%

Q43. Let us say that QC decides to sample one more tablet from Batch Y4. Batch Y4 has a mean paracetamol level of 505 mg and its standard deviation is 25 mg. What is the probability that the tablet selected by QC has a paracetamol level between 450 mg and 550 mg?

- 91%
- 93%
- 95% $\longrightarrow$ CORRECT
- 97%

Q44. The graph below represents the PDF of a uniformly distributed random variable X . The probability density is equal for all possible values of X (-5 to +5). What is the probability that the random variable X lies between -1.5 and +2.5, i.e., $P(-1.5 < X < 2.5)$?

- 0.1
- 0.4 $\longrightarrow$ CORRECT
- 4.0
- 0.6

(Gaussian Distribution)

The normal distribution was discovered by Carl Friedrich Gauss in 1809. Gauss was trying to create a probability distribution for astronomical errors. For example, Gauss found that an astronomer trying to estimate the distance between Earth and Uranus always makes an error. This error is normally distributed, with $\mu = 0$ km and $\sigma = 1,000$ km.

Q45. Based on the above information, what is the probability that the astronomer overestimated the distance by 2,330 km or more?

- 1% $\longrightarrow$ CORRECT
- 2%
- 1.5%
- 0.5%

Q46. What is the probability that the astronomer under- or over-estimates the distance by less than 500 km?

- 30.85%
- 69.15%
- 38.30% $\longrightarrow$ CORRECT
- 48.25%

C1 - M1 - S4 (Linear Transformations)

Q47. Which of the following matrices cannot be used for shear transformation? (Note: More than one option may be correct.)

- $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \longrightarrow$ CORRECT
- $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \longrightarrow$ CORRECT

Q48. Consider the transformation $T(x_1, x_2) = (5x_1, 2x_2)$. Which of the following is/are correct? (Note: More than one option may be correct.)

- $T(1, 2) = (5, 4) \longrightarrow$ CORRECT
- $T(1, -2) = (5, -4) \longrightarrow$ CORRECT
- $T(1, 1) = (5, 2) \longrightarrow$ CORRECT
- $T(1, -1) = (5, -2) \longrightarrow$ CORRECT
- $T(0, 2) = (0, 4) \longrightarrow$ CORRECT

Q49. Consider the transformation given by the matrix $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Which of the following statements is correct?

- The transformation is reversible
- The transformation is irreversible $\longrightarrow$ CORRECT

Q50. What are the coordinates of the vector $(3, 3)$ in the basis $\{(0, 1), (1, 1)\}$?

- $(1, 2)$
- $(1, 3)$
- $(0, 3) \longrightarrow$ CORRECT

Q51. Consider the basis $\{(1, -1), (1, 1)\}$. If the coordinates of a vector are $(2, 3)$ on this basis, then the vector is equal to?

- $(5, 2)$
- $(5, 1) \longrightarrow$ CORRECT
- $(5, 0)$

Question 52. Consider the basis $\{(0, 1), (1, 1)\}$. The matrix used for anti-clockwise rotation by 90 degrees in this basis is given by?

- $\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \longrightarrow$ CORRECT
- $\begin{bmatrix} 1 & -2 \\ -1 & -1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$

Q53. Use the matrix calculated in the previous question to find the coordinates of the vector $(3, 2)$ after being rotated by the matrix on the basis $\{(0, 1), (1, 1)\}$.

- $(7, 5)$
- $(7, -5) \rightarrow \text{CORRECT}$
- $(-7, -5)$

Q54. The transformation defined by $T(x_1, x_2) = (x_1^2, x_2)$ is?

- Non-Linear $\rightarrow \text{CORRECT}$
- Linear

Q55. Consider the transformation given by the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$. Which of the following is correct?

- The transformation is reversible
- The transformation is irreversible $\rightarrow \text{CORRECT}$

Q56. Which of the following is a matrix used for shear transformation on a standard basis?
(Note: More than one option may be correct.)

- $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \rightarrow \text{CORRECT}$
- $\begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$
- $\begin{bmatrix} 5 & 5 \\ 0 & 5 \end{bmatrix}$
- $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

Q57. Let T be a matrix transformation such that one of its eigenvectors is the vector V and its corresponding eigenvalue is E . Which of the following statements is necessarily true?

- If T is applied on V , the length of V changes.
- If T is applied on V , the direction of V changes.
- None of the above $\longrightarrow$ CORRECT

Q58. Consider the transformation defined by the matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$. The number of distinct eigenvalues of this transformation is equal to?

- 1
- 2 $\longrightarrow$ CORRECT
- 3

Q59. Which of the following is not an eigenvalue of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$?

- 2
- -5
- 4 $\longrightarrow$ CORRECT
- None of the above

Q60. Which of the following is not an eigenvalue of the matrix $\begin{bmatrix} 3 & -8 & 4 \\ 0 & -5 & -1 \\ 0 & 0 & 6 \end{bmatrix}$?

- 2 $\longrightarrow$ CORRECT
- 3
- -5
- 6

Q61. Consider the transformation $T(x_1, x_2) = (5x_1, 2x_2)$. Is the vector $(1, 0)$ an eigenvector of this transformation?

- Yes $\longrightarrow$ CORRECT
- No

Q62. Consider the transformation $T(\bar{x}) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \bar{x}$. The vector $(1, -1)$ is an eigenvector that corresponds to what eigenvalue?

- 2
- 1
- -1 $\longrightarrow$ CORRECT

Q63. Suppose that the eigen decomposition of a transformation T is given by $T = CDC^{-1}$. Which of the following statements is wrong?

- The matrix C contains the eigenvectors of the transformation T
- The matrix D contains the eigenvectors of the transformation T
- The matrix D is a diagonal matrix
- None of the above $\longrightarrow$ CORRECT

Q64. Suppose that the eigen decomposition of a matrix is given by $A = CDC^{-1}$. If $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ then matrix C is equal to?

- $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \longrightarrow$ CORRECT
- $\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Q65. Suppose that the eigen decomposition of a matrix is given by $A = CDC^{-1}$. If $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ then matrix D is equal to?

- $\begin{bmatrix} 5 & 0 \\ 0 & -3 \end{bmatrix}$
- $\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} \longrightarrow \text{CORRECT}$
- $\begin{bmatrix} -5 & 0 \\ 0 & 3 \end{bmatrix}$

Q66. Find the image of the origin under transformation $T(x_1, x_2) = (x_1 + 1, x_2 + 2)$.

- $(0, 0)$
- $(1, 5)$
- $(3, 2)$
- $(1, 2) \longrightarrow \text{CORRECT}$

Q67. Which of the following matrices can be used for dilation transformation in the standard basis?

- $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \longrightarrow \text{CORRECT}$
- $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Q68. Consider the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. This matrix transforms the vector $(1, 1)$ to the vector?

- $(1, 1)$
- $(0, 1)$
- $(3, 1)$ $\longrightarrow$ CORRECT
- None of the above

Q69. What are the coordinates of the vector $(1, 3)$ in the basis $\{(1, 0), (0, 1)\}$?

- $(1, 2)$
- $(3, 1)$
- $(1, 3)$ $\longrightarrow$ CORRECT
- $(2, 1)$

Q70. Consider the transformation $T(\bar{x}) = \begin{bmatrix} 2 & 2 \\ 5 & -1 \end{bmatrix} \bar{x}$. What is the eigenvector that corresponds to the eigenvalue -3 ?

- $(2, -5)$ $\longrightarrow$ CORRECT
- $(1, -5)$
- $(3, -5)$
- $(4, -5)$

Q71. Consider the transformation defined using the matrix $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. Which of the following matrices defines the transformation T applied thrice on a vector?

- $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$
- $\begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} \longrightarrow \text{CORRECT}$
- $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$
- $\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$

Q72. Suppose that the eigen decomposition of a matrix is given by $A = CDC^{-1}$. If $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ then matrix D is equal to?

- $\begin{bmatrix} -3 & 0 \\ 0 & -5 \end{bmatrix}$
- $\begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \longrightarrow \text{CORRECT}$
- $\begin{bmatrix} -1 & 1 \\ 0 & 5 \end{bmatrix}$

C1 - M1 - S5 (Multivariable Calculus)

Q73. What is the range of the function $z = x^2 + y^2$?

- $\mathbb{R}$
- All real numbers greater than or equal to zero $\longrightarrow$ CORRECT
- $\mathbb{R} - \{0\}$
- $[1, \infty]$

Q74. What is the domain of the function $f(x, y) = xy + e^x$?

- $\mathbb{R}^2 \longrightarrow$ CORRECT
- $\mathbb{R}^2 - \{(0, 0)\}$
- None of the above

Q75. Consider the function $f(x, y) = x^3 - x \cos y$. Then $\frac{\delta f}{\delta x} = ?$

- $3x^2 - x \sin y$
- $3x^2 - \cos y \longrightarrow$ CORRECT
- $x^3 - x \sin y$

Q76. Consider the function $f(x, y) = y \sin x - x \cos y$. Then, at $\left(\frac{\pi}{2}, \frac{-\pi}{2}\right)$ the value of $\frac{\delta f}{\delta y}$ is equal to?

- $1 - \frac{\pi}{2} \longrightarrow$ CORRECT
- $1 + \frac{\pi}{2}$
- $\frac{\pi}{2}$

Q77. For the function $f(x, y) = \sin(xy) - 3xy$, find the value of $\frac{\delta f}{\delta x}$ and $\frac{\delta f}{\delta y}$.

- $\frac{\delta}{\delta x}f(x, y) = y \cos(xy) - 3y$ and $\frac{\delta}{\delta y}f(x, y) = x \cos(xy) - 3x \longrightarrow \text{CORRECT}$
- $\frac{\delta}{\delta x}f(x, y) = y \cos(xy) + 3$ and $\frac{\delta}{\delta y}f(x, y) = x \cos(xy) + 3$
- $\frac{\delta}{\delta x}f(x, y) = y \cos(x) + 3y$ and $\frac{\delta}{\delta y}f(x, y) = x \cos(y) + 3x$

Q78. Given $f(x, y) = xy \ln(xy)$, find the value of $\frac{\delta f}{\delta x}$ and $\frac{\delta f}{\delta y}$.

- $y \ln(xy) + y$ and $x \ln(xy) + x$ respectively $\longrightarrow \text{CORRECT}$
- $\ln(xy) + y$ and $\ln(xy) + x$ respectively
- $y \ln(xy)$ and $x \ln(xy)$ respectively
- None of the above

Q79. Given $f(x, y, z) = x^2y + y^2z + z^2x$, what are $\frac{\delta f}{\delta x}$, $\frac{\delta f}{\delta y}$ and $\frac{\delta f}{\delta z}$ at $(1, 1, 1)$?

- $\frac{\delta f}{\delta x} = 1, \frac{\delta f}{\delta y} = 1, \frac{\delta f}{\delta z} = 1$
- $\frac{\delta f}{\delta x} = 0, \frac{\delta f}{\delta y} = 0, \frac{\delta f}{\delta z} = 0$
- $\frac{\delta f}{\delta x} = 3, \frac{\delta f}{\delta y} = 3, \frac{\delta f}{\delta z} = 3 \longrightarrow \text{CORRECT}$
- $\frac{\delta f}{\delta x} = 2, \frac{\delta f}{\delta y} = 2, \frac{\delta f}{\delta z} = 2$

Q80. Given that $f(x, y) = \pi^2 x^2 y$, $x(t) = t^2 + 1$ and $y(t) = t^2 - 1$, calculate the total derivative $\frac{df}{dt}$.

- $\frac{df}{dt} = 2\pi(t^2 + 1)^2(t^2 - 1) + \pi(t^2 + 1)^2(t^2 - 1)$
- $\frac{df}{dt} = 4\pi^2 t(t^2 + 1)(t^2 - 1) + 2\pi^2 t(t^2 + 1)^2 \longrightarrow \text{CORRECT}$
- $\frac{df}{dt} = 8\pi^2 t(t^2 + 1)^2(t^2 - 1)$
- $\frac{df}{dt} = 8\pi t(t^2 + 1)^2(t^2 - 1) + 2\pi t(t^2 + 1)^2$

Q81. Consider the function $f(x, y) = y \sin x - e^{-x}$. Then $\nabla f(x, y) = ?$

- $(y \sin x + e^{-x}, \sin x)$
- $(y \cos x + e^{-x}, \sin x) \longrightarrow \text{CORRECT}$
- $(y \cos x + e^{-x}, \cos x)$

Q82. Consider the function $f(x, y, z, w) = e^x - y - 3zw$. Which of the following is/are correct? (Note: More than one option may be correct.)

- $f(1, 3, -2, -1) = e - 9 \longrightarrow \text{CORRECT}$
- $f(1, 1, 1, 0) = e - 1 \longrightarrow \text{CORRECT}$
- $f(1, 3, -2, -1) = e - 8$
- $f(1, 1, 1, 0) = e$

Q83. What is the range of the function $z = -y^2 e^x$?

- $\mathbb{R}$
- The set of all real numbers less than or equal to zero $\longrightarrow \text{CORRECT}$
- $\mathbb{R} - \{0\}$

Q84. Consider the function $f(x, y) = ye^x - xy$. Then $\frac{\delta f}{\delta x} = ?$

- $ye^x - xy$
- $ye^x - y \longrightarrow \text{CORRECT}$
- $e^x - y$
- $e^x + y$

Q85. Consider the function $f(x, y) = \sin x \cos y$. Then $\frac{\delta f}{\delta y} = ?$

- $\cos x \sin y$
- $\sin x \cos y$
- $\sin x \sin y$
- None of the above (Answer: $-\sin x \sin y$) $\longrightarrow \text{CORRECT}$

Q86. Consider the function $f(x, y) = xy$. Then $\nabla f(x, y) = ?$

- (y, xy)
- $(y, x) \longrightarrow \text{CORRECT}$
- (xy, x)
- None of the above

Q87. Consider the function $f(x, y) = e^x + e^y$. Then, the direction of the steepest descent at the point $(1, 1)$ is given by the vector?

- $(2e, 2e)$
- $(-e, e)$
- $(-e, -e) \longrightarrow \text{CORRECT}$

C1 - M2 - S1 (Multivariable Calculus)

Q88. Which of the following fields can be a possible primary key of the **Marks** table?
Multiple options can be correct.

Marks

marks_range	grades
90–100	A
70–90	B

- grades → CORRECT
- marks_range → CORRECT
- There is no primary key

Q89. What is/are the foreign key(s) in the **Student** table? *Multiple options can be correct.*

Student

student_id	student_name	marks_range	year	branch_id
S-04	Ashish Mehra	50–60	1st	EE
S-05	Chetan Singh	70–90	3rd	ME

- student_id
- marks_range → CORRECT
- year
- branch_id → CORRECT

Q90. Choose the correct statements for the **HoD** table from the options given below. Multiple options can be correct.

Note: Assume that an HoD serves a branch only for a maximum of 1 year.

HoD

branch_HoD_name	year_of_service_as_HoD	branch_id	HoD_contact
Rakesh Sharma	2008	EE	9876543123
Vinay Rathod	2009	EE	9567456321
Rudra Singh	2010	CS	9876789044

- **branch_id** can be the primary key of the **HoD** table.
- **branch_id** is the foreign key of the **HoD** table. → CORRECT
- **branch_id** and **HoD_contact** can be the composite primary key of the **HoD** table. → CORRECT
- **branch_id** and **branch_HoD_name** can be the composite primary key of the **HoD** table.

Q91. Based on the tables which of the following statements could be correct?

Student

student_id	student_name	marks_range	year	branch_id
S-04	Ashish Mehra	50–60	1st	EE

Branch

branch_id	branch_name
EE	Electrical Engineering

HoD

branch_HoD_name	year_of_service_as_HoD	branch_id	HoD_contact
Rakesh Sharma	2008	EE	9876543123

Marks

marks_range	grades
90–100	A

- In the '**Branch**' table, 'branch_id' can have NULL values.
- In the '**Student**' table, 'marks_range' can have NULL values. → CORRECT
- In the '**Student**' table, 'branch_id' cannot have NULL values.
- In the '**HoD**' table, 'branch_id' can have NULL values.

Q92. In a star schema, what is the name of the table that contains the metadata that is needed for analyzing numerical/quantitative data?

- Fact Table
- Dimension Table → CORRECT
- All tables in the schema
- None of the above

Q93. Entity constraints, such as primary key, UNIQUE and NOT NULL are used to:

- Validate the quality of data entered for a specific property. → CORRECT
- Control who is allowed access to the data.
- Ensure that duplicate records are not entered into the table. → CORRECT
- Prevent users from changing the values stored in the table.

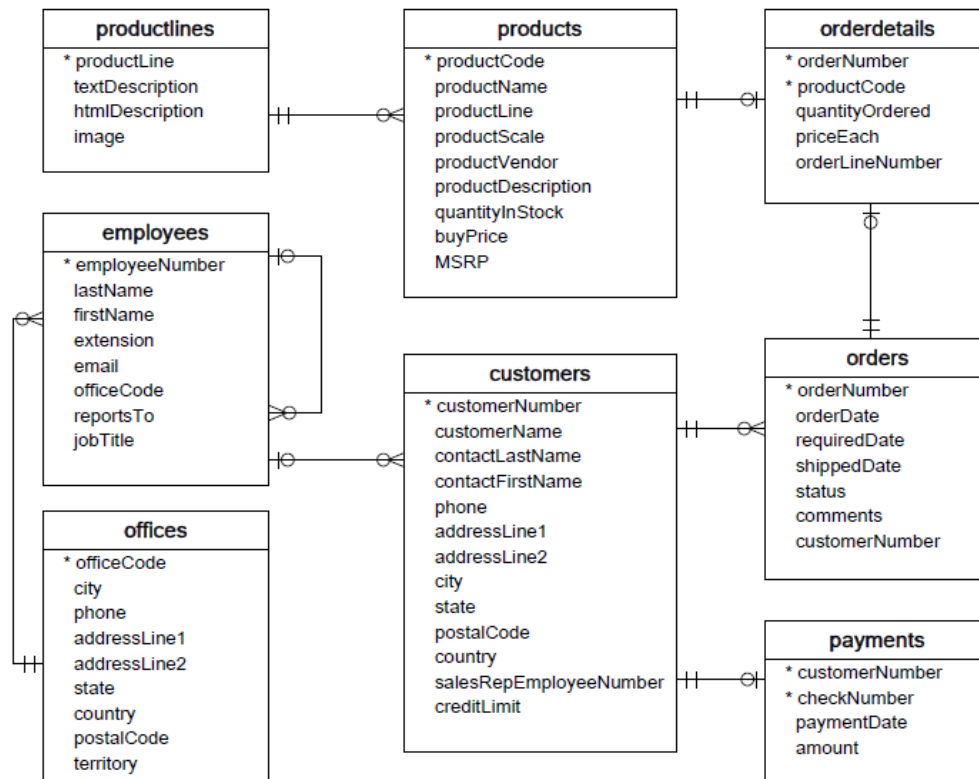
Q94. A primary key constraint always enforces both the UNIQUE and NOT NULL constraints.

- True → CORRECT
- False

Q95. Which of the following scenarios would require using a referential constraint such as a foreign key?

- All phone numbers should contain the area code.
- Certain fields are mandatory to fill (such as phone number) before the record is accepted into a database.
- A customer's details must be known before anything can be sold to them, so you can refer to the other fields to get information about the customer. → CORRECT
- When entering the card number for payment, the user must input a 16-digit number.

Q96. Refer to the ERD given above for the 'classic models' data set. Which of the following statements is/are correct?



- An employee is associated with zero or more customers, whereas a customer is associated with zero or one employee. → CORRECT
- A customer is associated with zero or more employees, whereas an employee is associated with zero or one customer.
- A customer can have zero or more orders, whereas an order is associated with one and only one customer. → CORRECT
- An order corresponds to zero or more customers, whereas a customer can have one and only one order.

Q97. How many fact variables are present in the **customers** table used in the following table?

Customers

customers
Customer_Name
City
State
Customer_Segment
Cust_id

Answer: 0

Since there are no numeric attributes in the 'cust_dimen' table, 0 is the correct answer. All its attributes are dimension variables.

Q98. Which type of variable is `Order_ID` in the **shipping** table?

shipping

Ship_id	VARCHAR(12)
Order_ID	INT(11)
Ship_Mode	VARCHAR(25)
Ship_Date	DATE

- Fact Variable
- Dimension Variable → CORRECT

C1 - M2 - S2 (Building blocks of Data modelling)

Q99. Why is data modelling necessary at the logical level of a database?

- To design different data elements in a particular structure. → CORRECT
- To design the physical structure in which data can be stored on devices.
- To give a specific structure to the data in accordance with the business requirements.
→ CORRECT
- To design applications that can be used to access data.

Q100. Which of the following elements does an E-R model identify real-life objects as?

- Entity → CORRECT
- Attribute
- Relation

Q101. Suppose that you are developing an E-R model for an online retail store, where every user has an 'Address'. The 'Address' attribute is divided into various other attributes such as City, State and Country. Among these attributes, only the values of the attribute Zip Code can be used to obtain the values of the attributes State and Country, as every city has a different zip code. Considering these factors, which of the following statements is correct?

- The address must be kept as an attribute.
- Zip Code must be kept as an attribute with address as a different entity. → CORRECT

Q102. Complete the following sentence with the correct option: In a relational model, an entity is a _____ ?

- Table → CORRECT
- Column
- Row
- Tuple

Q103. Suppose that you are an analyst in a bank and have a data file with the following six variables:

- (A) Withdrawal amount
- (B) Account balance after withdrawal
- (C) Transaction charge amount
- (D) Customer ID
- (E) ATM ID
- (F) Date of withdrawal

Based on a star schema, how many dimension variables would you need in order to determine the number of withdrawals made in Puducherry by a customer named Rohit Sharma?

- At least 3
- At least 1
- At least 4
- At least 2 $\longrightarrow$ CORRECT

Q104. You have the following tables and their columns:

1. **Table: student-grade**

This table contains the grades of every subject for all the students.

Columns: |*student_id*||*course_name*||*grade*|

2. **Table: polling**

This table contains the choices made in all polls by all users.

Columns: |*user_id*||*poll_id*||*choice*|

3. **Table: sales**

This table contains the number of products that a customer has ever purchased.

Columns: |*customer_id*||*product_id*||*product_count*|

Figure out, in each table, which attribute(s) should be considered as the key.

- (1) **student_grade**:|*student_id*|
(2) **polling**:|*user_id*|
(3) **sales**:|*customer_id*|
- (1) **student_grade**:|*student_id*|
(2) **polling**:|*user_id*|
(3) **sales**:|*customer_id*||*product_id*|
- (1) **student_grade**:|*student_id*|
(2) **polling**:|*user_id*||*poll_id*|
(3) **sales**:|*customer_id*||*product_id*|
- (1) **student_grade**:|*student_id*||*course_name*|
(2) **polling**:|*user_id*||*poll_id*| $\longrightarrow$ CORRECT
(3) **sales**:|*customer_id*||*product_id*|

Q105. Which of the following statements is correct regarding keys in DBMS?

Note: More than one option may be correct:

- The primary key must be unique for a relation $\longrightarrow$ CORRECT
- A candidate key is the largest possible set of attributes that can uniquely identify a record, and there can be multiple candidate keys in a table.
- A foreign key can reference a non-primary key column in another table, regardless of whether it is unique or not.
- A composite key can be used only when a single column is insufficient. $\longrightarrow$ CORRECT

Q106. Consider three different entities - **Department**, **Employee** and **User**.

The attributes for these entities are given along with them in the options. Which of the following entities has the incorrect set of attributes?

- **Department** - Name, Manager and Number_of_Employees
- **Employee** - Employee_ID and Name
- **User** - Login_ID, Name and Product_ID → CORRECT

Q107. Consider the database of a company having an ‘employee’ table containing the details of each employee and a ‘department’ table containing the data of different departments in the company. What kind of relationship do you think exists between the ‘department’ table and the ‘employee’ table?

- one-to-one relationship
- one-to-many relationship → CORRECT
- many-to-many relationship
- These two tables are not related

Q108. How does a table in a relational model provide data quality assurance?

Note: Multiple options may be correct.

- By ensuring that non-null fields contain values → CORRECT
- By ensuring that every field contains only permitted values → CORRECT
- By ensuring that the data in a field is of the specified data type → CORRECT
- By ensuring that the rows are in a particular order

Q109. Suppose for one row in the entity Team, there is only one corresponding row in the entity Project. And for one row in the entity Project, there are many corresponding rows in the entity Team. Which of the following relations exists from the entity Team to the entity Project in this case?

- A one-to-one relation
- A one-to-many relation
- A many-to-one relation → CORRECT
- A many-to-many relation

Q110. Entity A and entity B have a many-to-many relation. Which of the following statements is correct?

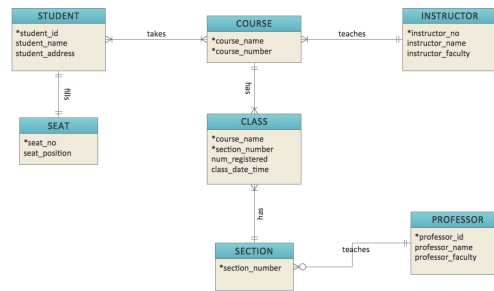
- Each row of entity A is necessarily related to many rows of entity B.
- Each row of entity B can be related to many rows of entity A. $\longrightarrow$ CORRECT
- Each row of entity A can be related to only one row of entity B.
- Each row of entity B is necessarily related to many rows of entity A.

Q111. Arrange the following in the correct order of the data flow in an organization to make a decision.

- (A) Storing data in the warehouse
- (B) Presenting a data-driven story using data visualization
- (C) Collecting data from data sources such as mobile applications
- (D) Modifying the extracted data from different sources to a standard form
- (E) Analyzing and discovering patterns in the historical data of an organization

- A $\longrightarrow$ B $\longrightarrow$ C $\longrightarrow$ D $\longrightarrow$ E
- C $\longrightarrow$ D $\longrightarrow$ A $\longrightarrow$ E $\longrightarrow$ B $\longrightarrow$ CORRECT
- B $\longrightarrow$ D $\longrightarrow$ C $\longrightarrow$ A $\longrightarrow$ E
- E $\longrightarrow$ D $\longrightarrow$ A $\longrightarrow$ C $\longrightarrow$ V

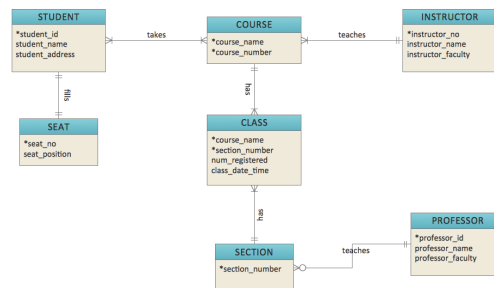
Q112. Based on the ERD given below :



How many tables in the ERD contain composite keys?

- 0
- 1
- 2 → CORRECT
- More than 2

Q113. Based on the ERD given below :



Which of the following statements is/are true? More than one option may be correct.

- The columns 'course_name' and 'section_number' in the table 'Class' are primary and foreign keys. → CORRECT
- A particular class can have multiple courses.
- A professor may teach none, one, or more sections. → CORRECT
- During schema creation, the 'Section' table must be created before the 'Class' table. → CORRECT

Q114. Consider the following statements:

- (A) CREATE
- (B) INSERT
- (C) UPDATE
- (D) DELETE
- (E) ALTER
- (F) DROP
- (G) TRUNCATE
- (H) RENAME

Based on the types of DDL/DML statements, select the correct options from below:

Note: Multiple options may be correct.

- CREATE, UPDATE, DELETE, DROP are DDL statements.
- SELECT, UPDATE, ALTER, DROP, RENAME are DML statements.
- CREATE, ALTER, DROP, RENAME, TRUNCATE Are DDL statements. →
CORRECT
- INSERT, UPDATE, DELETE are DML statements. → CORRECT

Q115. What is the difference between the 'DROP' and 'TRUNCATE' commands?

- 'TRUNCATE' removes the table completely from the database, and you can not retrieve this table. All the integrity constraint is removed. On the other hand, 'DROP' only drops the table rows; it does not delete the schema of the table from the database.
- 'DROP' removes the table completely from the database, and you can not retrieve this table. All the integrity constraint is removed. On the other hand, 'TRUNCATE' only drops the table rows; it does not delete the schema of the table from the database. → CORRECT
- The table remains in the database when you use the 'DROP' command, but all of its rows will vanish.
- The table is deleted completely from the database when you use the 'TRUNCATE' command. If you want to get this table, you need to create the table again from scratch using the 'create table' command.

Q116. Suppose that you want to create a table named 'mobile_phone' which contains the following columns:

- (A) **Mobile_phone_ID**, which is an integer type
- (B) **Camera**, which is a string type
- (C) **RAM**, which is an integer type
- (D) **Memory**, which is an integer type

The primary key in this table is '**Mobile_phone_ID**'.

What will be the right SQL query to create such a table? Also, identify the type of operation, i.e., whether it is a DDL statement or a DML statement.

- Its a DML statement

```
CREATE TABLE mobile_phone (  
  Mobile_phone_ID INT(11) NOT NULL,  
  Camera VARCHAR(12),  
  RAM INT(11),  
  Memory VARCHAR(12)  
);
```

- Its a DDL statement

```
CREATE TABLE mobile_phone (  
  Mobile_phone_ID INT(11) NOT NULL,  
  Camera VARCHAR(12),  
  RAM INT(11),  
  Memory VARCHAR(12)  
);
```

- Its a DML statement

```
CREATE TABLE mobile_phone (  
  Mobile_phone_ID INT(11) NOT NULL,  
  Camera VARCHAR(12),  
  RAM INT(11),  
  Memory VARCHAR(12)  
  Primary Key(Mobile_phone_ID)  
);
```

- Its a DDL statement

```
CREATE TABLE mobile_phone (
  Mobile_phone_ID INT(11) NOT NULL,
  Camera VARCHAR(12),
  RAM INT(11),
  Memory VARCHAR(12)
  Primary Key(Mobile_phone_ID)
);
```

→ CORRECT

Q117. Suppose you want to add a new column: '**price**', with 'float' as the data type, in the table '**mobile_phone**'. What is the right query to add this new column?

- ALTER 'mobile_phone' ADD COLUMN 'price' float(24)
- ADD COLUMN 'price' float(24) IN TABLE 'mobile_phone'
- ALTER TABLE 'mobile_phone' ADD COLUMN 'price' float(24) → CORRECT
- ADD COLUMN 'price' float(24) IN 'mobile_phone'

Q118. Suppose you want to update the value of 'Camera' to 15MP, where the RAM = 8. What would be the right query to perform this operation?

- update mobile_phone
set Camera= '15MP'
where RAM= 8;
→ CORRECT
- alter mobile_phone
set Camera= '15MP'
where RAM= 8;
- insert Camera= '15MP'
where RAM= 8;
in table mobile_phone
- update mobile_phone
set Camera= '15MP'
where RAM= '8';

Q119.What will be the output of the following query?

```
select * from employee
where first_name like ' '
```

- It will display the details of all employees whose names contain 'ee'. → CORRECT
- It will display the details of all employees whose names both begin and end with 'e'.
- It will display the details of all employees whose names only begin with 'ee'.
- It will display the details of all employees whose names only end with 'ee'.

Q120.Refer to the [MySQL documentation](#) online and answer the following question:

Which of the following wildcards are supported in MySQL with the LIKE operator?

- % → CORRECT
- _ → CORRECT
- *
- ^

Q121.With SQL, how do you select all the records from a table named 'employees' where the 'LastName' is alphabetically between (and includes) 'Joshi' and 'Sharma'?

- SELECT * from employees WHERE LastName > 'Joshi' and LastName < 'Sharma';
- SELECT * from employees WHERE LastName BETWEEN 'Joshi' and 'Sharma';
→ CORRECT
- Both of the above
- None of the above

Q122.Which of the queries given below would return the same output? More than one option may be correct.

- SELECT * FROM employees WHERE EmpID in (2,3,4, 5); → CORRECT
- SELECT * FROM employees WHERE EmpID BETWEEN 2 and 5; → CORRECT
- SELECT * FROM employees WHERE EmpID >= 2 and EmpID <= 5; → CORRECT
- SELECT * FROM employees WHERE EmpID > 2 and EmpID < 5;

Q123.What will be the output of the following query:

```
SELECT e_name
FROM employee
WHERE e_department != 'D3';
```

- e_name of all records in the employee table.
- e_name of all records in the employee table, where e_department is equal to D3.
- e_name of all records in the employee table, where e_department is not equal to D3.
→ CORRECT
- All the attributes of all records of the employee table.

Q124.What will be the output of the following query:

```
SELECT *
FROM employee
WHERE sex = 'F'
OR
salary <= 25000;
```

- All female employees whose salary is less than or equal to 25,000.
- All employees whose salary is less than or equal to 25,000, these employees may be male or female.
- All male employees whose salary is less than or equal to 25,000.
- All employees whose salary is less than or equal to 25,000 or all the female employees.
→ CORRECT

Q125.What will be the output of the following query:

```
SELECT *
FROM employee
WHERE e_department
IN ('D1', 'D3');
```

- All employees present in the employee table.
- All employees who belong to departments other than D1 and D3.
- All employees who belong to departments D1 or D3. → CORRECT
- Nothing will be displayed.

Q126. If you do not specify either of the 'asc' or 'desc' commands after the 'orderby' command, what will the output of this query be?

- This will throw an error.
- By default, it arranges the data in ascending order. → CORRECT
- By default, it arranges the data in descending order.

Q127. Suppose that you are given the following table named 'Student'.

student_id	student_name	grades	year	branch_id
S-01	Ashish Gupta	A	1	EE
S-02	Vinay Singh	D	4	ME
S-03	Chetan Verma	F	3	ECE
S-04	Arpit Mohan	B	2	CS

Select the right query to arrange students in increasing order of their 'year', and within the year, they should be arranged in increasing order of their grades.

- SELECT *
FROM Student
ORDER BY year, grades; → CORRECT
- SELECT *
FROM Student
ORDER BY grades, year;
- SELECT *
FROM Student
ORDER BY year
ORDER BY grades;
- SELECT *
FROM Student
ORDER BY grades
ORDER BY year;

Q128. Note: Use the movie database for this question.

What is the average movie duration by language, for movies in the movie_main dataset?

- Korean - 110, English - 110.83, Japanese - 127.67, Hungarian - 134 → CORRECT
- Korean - 110, English - 110.83, Hungarian - 127.67, Japanese - 134
- Hungarian - 110, English - 110.83, Japanese - 127.67, Korean - 134
- Korean - 110, Japanese - 110.83, English - 127.67, Hungarian - 134

Q129. Which counties are the top 3 counties in terms of the number of collisions?

Note - Use the collisions database for this question. Don't forget to run **USE collisions;** in your SQL coding environment to update your default schema to **collisions**.

- Los Angeles, Santa Clara, Riverside
- Los Angeles, Orange, San Diego
- Los Angeles, Orange, San Bernandino → CORRECT
- Los Angeles, Alameda, San Diego

Q130. What will be the output of the following query?

```
SELECT id, col_1, col_2
  FROM table_1 a
    inner join table_2 b
      using(id)
    inner join table_3 c
      using(id_2);
```

Note: Assume id and id_2 are present in all tables

- This query will show an error message.
- This query will perform the inner join of table_1 and table_3 only.
- This query will perform the inner join of table_1 and table_2 on 'id' and then perform inner join on 'id_2' with table_3. → CORRECT
- This query will perform the inner join of table_2 and table_3 on 'id_2' and then perform inner join on 'id' with table_1.

Q131. The left and right joins are also known as _____ ?

- Inner Join
- Natural Join
- Outer join → CORRECT
- Cartesian join

Q132. You have been given the following two tables: **Transactions** and **Company**.

Transactions

Check_no	Company_id	Date	Amount
SBI1234	B	18/01/2020	19877
AXIS2476	D	23/05/2020	20876
HDFC7612	B	09/09/2020	45261
ICICI1010	E	12-12-2020	10298

Company

Company_id	Company_name
A	Hyundai
B	Toyota
C	Tata
E	Ford
F	Maruti

If you perform an **INNER JOIN** on these two tables using **Company_id**, how many rows will the resulting table contain?

- 4
- 3 → CORRECT
- 2
- 5

Q133. Every SQL statement containing a 'join' keyword can be rewritten with a 'using' keyword.

- True
- False $\longrightarrow$ CORRECT

Q134. Match the following actions with the appropriate SQL commands.

a	Find all the employees from the employees table	I	ALTER
b	Add information about a new employee	II	DROP
c	Change the structure of the employee table	III	SELECT
d	Remove the employees table	IV	INSERT

- a-I, b-II, c-III, d-IV
- a-IV, b-III, c-II, d-I
- a-III, b-I, c-II, d-IV
- a-III, b-IV, c-I, d-II $\longrightarrow$ CORRECT

Q135. Which statement is used to count the number of rows in a table?

- SELECT COUNT(*) FROM table_name; $\longrightarrow$ CORRECT
- SELECT COUNT ALL(*) FROM table_name;
- SELECT ROWS(*) FROM table_name;
- Both A and B

Q136. Suppose that you are given a table named 'Customers'. There are two columns in the table – 'firstName' and 'lastName'. If one of the entries in the table is as follows – **firstName: Diane & lastName: Murphy**

What will be the output of the following set of commands?

```
SELECT CONCAT (REVERSE(firstName), ' ', UPPER(lastName))  
FROM Customers  
WHERE customer_id= 'C-05';
```

- Diane MURPHY
- Enaid Murphy
- enaiD MURPHY → CORRECT
- enaiDMURPHY

Q137. List the given MySQL keywords in the order of their occurrence in a single query.

- (a) group by
- (b) order by
- (c) select
- (d) where
- (e) from
- (f) limit
- (g) having

- 3 - 5 - 4 - 1 - 2 - 7 - 6
- 3 - 5 - 1 - 4 - 2 - 6 - 7
- 3 - 5 - 4 - 1 - 7 - 2 - 6 → CORRECT
- 5 - 4 - 1 - 7 - 3 - 2 - 6

Q138. You have been given the following three tables: **Transactions**, **Company**, and **Headquarter**.

Transactions

Check_no	Company_Headquarter_id	Date	Amount
SBI1234	HQ01	18/01/2020	19877
AXIS2476	HQ02	23/05/2020	20876
HDFC7612	HQ03	09/09/2020	45261
ICICI1010	HQ04	12-12-2020	10298

Headquarter

Company_Headquarter_id	Company_id
HQ01	A
HQ02	B
HQ03	C
HQ04	D
HQ-5	E

Suppose you want to get the company name corresponding to the highest amount paid in the ‘Transactions’ table. Which method will be the most time-efficient for writing queries to perform such an operation?

- Perform the operation in 3 steps:
 - First, get the ‘Company_Headquarter_id’ corresponding to the highest amount from the ‘Transactions’ table.
 - Then, find the ‘Company_id’ from the ‘Headquarter’ table that corresponds to the ‘Company_headquarter_id’ obtained from the above step.
 - Perform the operation on the ‘Company’ table to get the ‘Company_name’.
- Perform the nested query operation among the three tables.
- Perform the operation on the ‘Company’ table to get the ‘Company_name’. →
CORRECT

Q139. Suppose there is an 'employee' table. You want to see what a 2% increment to all the employees having 'e_department' as 'D1' will look like. Which of the following options is the correct query for this?

- `SELECT salary + 0.2*salary
FROM employee
WHERE e_department = 'D1';` → CORRECT
- `SELECT salary + 0.2*salary
FROM employee`
- `SELECT 0.2*salary
FROM employee
WHERE e_department = 'D1';`
- `SELECT 0.2*salary
FROM employee;`

Q140. Which of the following statements is NOT true for views in SQL?

- Select statements used in the view definition cannot include ORDER BY clause.
- A view derives its data from the base tables(s).
- A view is updatable if it has been defined from a single relation.
- A view contains a copy of the data. → CORRECT

Q141. Fill in the blank.

If A and B are two sets, then $A \cap (B \cup A)$ equals ----.

- A → CORRECT
- B
- A - B
- None of the above

Q142. Consider the two tables below:

Employees

Last_Name	Dept_ID	Salary
Getz	10	3000
Davis	20	1500
King	20	2200
Davis	30	5000
Kochhar		5000

Department

Dept_ID	Dept_Name
10	Sales
20	Marketing
30	Accounts
40	Administration

Which query would you write to retrieve all employees, whether or not they have matching departments in the **Departments** table?

- ```
select last_name, dept_name
from employees e
left join departments d
on e.dept_id = d.dept_id;
```

↔ CORRECT

- ```
select last_name, dept_name
from employees e
right join departments d
on e.dept_id = d.dept_id;
```
- ```
select last_name, dept_name
from employees e
inner join departments d
on e.dept_id = d.dept_id;
```
- None of the above

**Q143.** Suppose you have been given the following two tables ‘Customers’ and ‘Orders’.

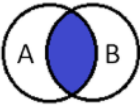
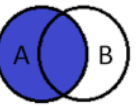
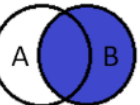
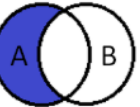
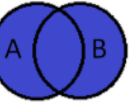
**Customers**

| project_no | project_type     | dept_no |
|------------|------------------|---------|
| P-100      | Machine Learning | D01     |
| P-101      | Data Analytics   | C01     |
| P-102      | Management       | E01     |
| P-103      | Cloud            | D01     |
| P-104      | Big Data         | B01     |

**Orders**

| dept_no | dept_name  |
|---------|------------|
| A01     | Sales      |
| B01     | HR         |
| C01     | Marketing  |
| D01     | Technology |

These two tables are to be joined on the 'dept\_no' key

|  <p>A: INNER JOIN</p>                | <table> <tr> <th>project_no</th><th>project_type</th><th>dept_no</th><th>dept_name</th></tr> <tr> <td>NULL</td><td>NULL</td><td>A01</td><td>Sales</td></tr> <tr> <td>P-104</td><td>Big Data</td><td>B01</td><td>HR</td></tr> <tr> <td>P-101</td><td>Data Analytics</td><td>C01</td><td>Marketing</td></tr> <tr> <td>P-100</td><td>Machine Learning</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-103</td><td>Cloud</td><td>D01</td><td>Technology</td></tr> </table> <p>I</p>                                                                       | project_no | project_type | dept_no | dept_name | NULL  | NULL             | A01 | Sales      | P-104 | Big Data       | B01 | HR        | P-101 | Data Analytics | C01 | Marketing  | P-100 | Machine Learning | D01 | Technology | P-103 | Cloud    | D01 | Technology |       |            |     |      |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|---------|-----------|-------|------------------|-----|------------|-------|----------------|-----|-----------|-------|----------------|-----|------------|-------|------------------|-----|------------|-------|----------|-----|------------|-------|------------|-----|------|
| project_no                                                                                                            | project_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | dept_no    | dept_name    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| NULL                                                                                                                  | NULL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | A01        | Sales        |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-104                                                                                                                 | Big Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | B01        | HR           |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-101                                                                                                                 | Data Analytics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C01        | Marketing    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-100                                                                                                                 | Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-103                                                                                                                 | Cloud                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
|  <p>B: LEFT OUTER JOIN</p>         | <table> <tr> <th>project_no</th><th>project_type</th><th>dept_no</th><th>dept_name</th></tr> <tr> <td>P-102</td><td>Management</td><td>E01</td><td>NULL</td></tr> </table> <p>II</p>                                                                                                                                                                                                                                                                                                                                                                      | project_no | project_type | dept_no | dept_name | P-102 | Management       | E01 | NULL       |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| project_no                                                                                                            | project_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | dept_no    | dept_name    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-102                                                                                                                 | Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | E01        | NULL         |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
|  <p>C: RIGHT OUTER JOIN</p>        | <table> <tr> <th>project_no</th><th>project_type</th><th>dept_no</th><th>dept_name</th></tr> <tr> <td>P-100</td><td>Machine Learning</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-101</td><td>Data Analytics</td><td>C01</td><td>Marketing</td></tr> <tr> <td>P-102</td><td>Management</td><td>E01</td><td>NULL</td></tr> <tr> <td>P-103</td><td>Cloud</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-104</td><td>Big Data</td><td>B01</td><td>HR</td></tr> </table> <p>III</p>                                                               | project_no | project_type | dept_no | dept_name | P-100 | Machine Learning | D01 | Technology | P-101 | Data Analytics | C01 | Marketing | P-102 | Management     | E01 | NULL       | P-103 | Cloud            | D01 | Technology | P-104 | Big Data | B01 | HR         |       |            |     |      |
| project_no                                                                                                            | project_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | dept_no    | dept_name    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-100                                                                                                                 | Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-101                                                                                                                 | Data Analytics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C01        | Marketing    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-102                                                                                                                 | Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | E01        | NULL         |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-103                                                                                                                 | Cloud                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-104                                                                                                                 | Big Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | B01        | HR           |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
|  <p>D: LEFT OUTER JOIN, ONLY A</p> | <table> <tr> <th>project_no</th><th>project_type</th><th>dept_no</th><th>dept_name</th></tr> <tr> <td>NULL</td><td>NULL</td><td>A01</td><td>Sales</td></tr> <tr> <td>P-104</td><td>Big Data</td><td>B01</td><td>HR</td></tr> <tr> <td>P-101</td><td>Data Analytics</td><td>C01</td><td>Marketing</td></tr> <tr> <td>P-100</td><td>Machine Learning</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-103</td><td>Cloud</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-102</td><td>Management</td><td>E01</td><td>NULL</td></tr> </table> <p>IV</p> | project_no | project_type | dept_no | dept_name | NULL  | NULL             | A01 | Sales      | P-104 | Big Data       | B01 | HR        | P-101 | Data Analytics | C01 | Marketing  | P-100 | Machine Learning | D01 | Technology | P-103 | Cloud    | D01 | Technology | P-102 | Management | E01 | NULL |
| project_no                                                                                                            | project_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | dept_no    | dept_name    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| NULL                                                                                                                  | NULL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | A01        | Sales        |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-104                                                                                                                 | Big Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | B01        | HR           |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-101                                                                                                                 | Data Analytics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C01        | Marketing    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-100                                                                                                                 | Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-103                                                                                                                 | Cloud                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-102                                                                                                                 | Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | E01        | NULL         |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
|  <p>E: FULL OUTER JOIN</p>         | <table> <tr> <th>project_no</th><th>project_type</th><th>dept_no</th><th>dept_name</th></tr> <tr> <td>P-100</td><td>Machine Learning</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-101</td><td>Data Analytics</td><td>C01</td><td>Marketing</td></tr> <tr> <td>P-103</td><td>Cloud</td><td>D01</td><td>Technology</td></tr> <tr> <td>P-104</td><td>Big Data</td><td>B01</td><td>HR</td></tr> </table> <p>V</p>                                                                                                                                      | project_no | project_type | dept_no | dept_name | P-100 | Machine Learning | D01 | Technology | P-101 | Data Analytics | C01 | Marketing | P-103 | Cloud          | D01 | Technology | P-104 | Big Data         | B01 | HR         |       |          |     |            |       |            |     |      |
| project_no                                                                                                            | project_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | dept_no    | dept_name    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-100                                                                                                                 | Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-101                                                                                                                 | Data Analytics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C01        | Marketing    |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-103                                                                                                                 | Cloud                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D01        | Technology   |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |
| P-104                                                                                                                 | Big Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | B01        | HR           |         |           |       |                  |     |            |       |                |     |           |       |                |     |            |       |                  |     |            |       |          |     |            |       |            |     |      |

- A-II, B-I, C-IV, D-V, E-III
- A-I, B-III, C-II, D-V, E-IV
- A-IV, B-III, C-I, D-II, E-V
- A-V, B-III, C-I, D-II, E-IV → CORRECT

**Q144.** Given below is a table containing the top 10 movies according to their IMDb ratings.

**IMDb Ratings**

| Title                                             | IMDb Rating | Dense Rank | Rank |
|---------------------------------------------------|-------------|------------|------|
| The Shawshank Redemption                          | 9.2         | 1          | 1    |
| The Godfather                                     | 9.1         | 2          | 2    |
| The Godfather: Part II                            | 9.0         | 3          | 3    |
| The Dark Knight                                   | 9.0         | 3          | 3    |
| 12 Angry Men                                      | 8.9         | 4          | 5    |
| Schindler's List                                  | 8.9         | 4          | 5    |
| The Lord of the Rings: The Return of the King     | 8.9         | 4          | 5    |
| Pulp Fiction                                      | 8.9         | 4          | 5    |
| The Good, the Bad and the Ugly                    | 8.8         | A          | B    |
| The Lord of the Rings: The Fellowship of the Ring | 8.8         | A          | B    |

Calculate the values (A,B).

- (5,6)
- (5,9) → CORRECT
- (6,6)
- (6,9)

**Q145.** Which of the following statements is true for window functions? More than one option may be correct.

- They compute a value for each and every row in the window. → CORRECT
- They reduce the result set based on the aggregation.
- They require the 'over' clause. → CORRECT
- Window partitions can be defined to form groups of rows. → CORRECT

**Q146.** Consider the following table with 11 rows.

|        |
|--------|
| Row 1  |
| Row 2  |
| Row 3  |
| Row 4  |
| Row 5  |
| Row 6  |
| Row 7  |
| Row 8  |
| Row 9  |
| Row 10 |
| Row 11 |

What is the range of rows that will be selected if you write the following frame clause?  
(Assume that the current row is Row 6)

ORDER BY X ROWS 4 FOLLOWING

- Row 6 to Row 9
- Row 6 to Row 10 → CORRECT
- Row 6 to Row 3
- Row 5 to Row 2

**Q147.** Consider the following table with 11 rows.

|        |
|--------|
| Row 1  |
| Row 2  |
| Row 3  |
| Row 4  |
| Row 5  |
| Row 6  |
| Row 7  |
| Row 8  |
| Row 9  |
| Row 10 |
| Row 11 |

What is the range of rows that will be selected if you write the following frame clause?  
(Assume that the current row is Row 6)

ORDER BY X ROWS  
BETWEEN 2 PRECEDING AND 3 FOLLOWING

- Row 3 to Row 8
- Row 4 to Row 8
- Row 4 to Row 9  $\longrightarrow$  CORRECT
- Row 5 to Row 9

**Q148.** Consider the following table with 11 rows.

|        |
|--------|
| Row 1  |
| Row 2  |
| Row 3  |
| Row 4  |
| Row 5  |
| Row 6  |
| Row 7  |
| Row 8  |
| Row 9  |
| Row 10 |
| Row 11 |

What is the range of rows that will be selected if you write the following frame clause?  
(Assume that the current row is Row 6)

ORDER BY X DESC ROWS  
BETWEEN UNBOUNDED PRECEDING AND 2 FOLLOWING

- Row 1 to Row 8
- Row 4 to Row 11 → CORRECT
- Row 1 to Row 11
- Row 1 to Row 7

**Q149.** Consider a table named 'marks', which has nine columns, as shown below:

**name:** Name of the student

**semester\_1:** Average marks in the first semester

**semester\_2:** Average marks in the second semester

**semester\_3:** Average marks in the third semester

**semester\_4:** Average marks in the fourth semester

**semester\_5:** Average marks in the fifth semester

**semester\_6:** Average marks in the sixth semester

**semester\_7:** Average marks in the seventh semester

**semester\_8:** Average marks in the eighth semester

What will be the output of the SQL query mentioned below?

```
SELECT name,
(CASE
 WHEN semester_1 > semester_2 THEN 'Sem 1'
 ELSE 'Sem 2'
END) AS 'Sem 1/2'
FROM marks;
```

- The names of the students and the semester out of 1 and 2 in which a student scored more marks. → CORRECT
- The names of the students, and 'Sem 2'.
- All student details and the semester out of 1 and 2 in which a student scored more marks.
- All student details, and 'Sem 2'.

**Q150.** Consider a table named “marks,” which has three columns, as shown below:

**name:** Name of the student

**SAT\_1:** Marks in the first attempt of the SAT exam

**SAT\_2:** Marks in the second attempt of the SAT exam

What will be the output of the SQL query mentioned below?

```
SELECT name ,
(CASE
 WHEN SAT_1 > SAT THEN 'SAT_1'
 ELSE 'SAT_2'
 END) AS case_column
FROM marks;
```

- The names of the students and the name of the SAT attempt (out of 'SAT\_1' and 'SAT\_2') in which the student scored more marks. → CORRECT
- The names of the students, and 'SAT\_2'.
- All student details and the name of the SAT attempt (out of 'SAT\_1' and 'SAT\_2') in which the student scored more marks.
- All student details, and 'SAT\_2'.

**Q151.** Suppose you have the following view that is built on the collisions dataset:

```
CREATE OR REPLACE VIEW num_parties_injured_by_collision_year AS (
 SELECT
 YEAR(collision_date) AS collision_year ,
 SUM(injured_victims) AS num_parties_injured
 FROM collisions
 GROUP BY collision_year
 ORDER BY collision_year
);
```

Querying all rows of this dataset gives you the number of parties injured in collisions every year you have data for.

What will be the SQL query to output columns **collision\_year**, **num\_parties\_injured**, and **num\_parties\_injured\_so\_far** using this view?

The column **num\_parties\_injured\_so\_far** is a running total of the number of parties injured up to the current year.

For example,

| collision_year | num_parties_injured | num_parties_injured_so_far |
|----------------|---------------------|----------------------------|
| 2001           | 2555                | 2555                       |
| 2002           | 3927                | 6482                       |
| 2003           | 3812                | 10294                      |

- SELECT  
\*,  
SUM(num\_parties\_injured) OVER (PARTITION BY collision\_year)  
AS num\_parties\_injured\_so\_far  
FROM num\_parties\_injured\_by\_collision\_year
- SELECT  
\*,  
SUM(num\_parties\_injured) OVER (ORDER BY collision\_year)  
AS num\_parties\_injured\_so\_far  
FROM num\_parties\_injured\_by\_collision\_year  
→ CORRECT
- SELECT  
\*,  
SUM(num\_parties\_injured) OVER ()  
AS num\_parties\_injured\_so\_far  
FROM num\_parties\_injured\_by\_collision\_year

**Q152.** Which of the following can be used to create or remove an index? More than one option can be correct.

- CREATE INDEX index\_name ON table\_name (column1, column2,...); → CORRECT
- ADD INDEX ON table\_name (column1, column2,...);
- ALTER TABLE table\_name DROP INDEX index\_name; → CORRECT
- REMOVE INDEX index\_name;

**Q153.** How many values can be returned from a given stored function?

- 0
- 1 → CORRECT
- 2
- More than 2

**Q154.** Which of the following can use a CTE as well as a TRY-CATCH block?

- UDFs
- Stored Procedures → CORRECT
- Both
- None of the above

**Q155.** Which of the following SQL clause works efficiently when you are using indices?

- SELECT
- FROM
- DROP
- WHERE → CORRECT

**Q156.** Arrange the following SQL statements in the correct order in which they are executed by the database engine.

- (a) SELECT
- (b) DISTINCT
- (c) FROM(including joins)
- (d) WHERE
- (e) GROUP BY
- (f) HAVING
- (g) Window functions (h) ORDER BY
- (i) LIMIT and OFFSET

- 1-2-3-4-5-6-7-8-9
- 1-2-7-3-4-5-6-8-9
- 3-4-5-6-7-1-2-8-9 → CORRECT
- 3-4-5-6-1-2-7-8-9

**Q157.** Instead of using DISTINCT, using the UNION operator is a more optimized approach. So, do you think the following query would be able to fetch German cities (only distinct values) from both the ‘Customers’ and ‘Suppliers’ tables?

```
SELECT City , Country FROM Customers
WHERE Country='Germany'
UNION
SELECT City , Country FROM Suppliers
WHERE Country='Germany'
ORDER BY City ;
```

- Yes → CORRECT
- No
- Error - ORDER BY clause cannot be used along with UNION operator
- Depends on data

**Q158.** Which statement holds good for a clustered index?

- It defines the order of the table data based on the key values that can be sorted in only one direction. → CORRECT
- In a database, each table can contain any number of clustered indexes.
- In a relational database, if the table column contains a primary key or unique key, then there is still a need to create an index on such a table.
- It has slower accessing power in comparison to the non-clustered index.

**Q159.** What does horizontal scaling mean in terms of scaling databases?

- Increasing the capacity and power of a single machine that stores the database by adding CPU, RAM, etc.
- Increasing the number of servers on which the database gets distributed in order to improve performance. → CORRECT
- Only increasing the CPU of the single machine on which the database is stored.

**Q160.** Designing a database having a fixed schema is possible in NoSQL Databases. True or False?

- True → CORRECT
- False

**Q161.** Fill in the blanks

Each \_\_\_\_ in a MongoDB database corresponds to a table in SQL, whereas every \_\_\_\_ corresponds to a record in the table in SQL databases.

- Database, Collection
- Document, Collection
- Collection, Database
- Collection, Document → CORRECT

**Q162.** Let's say you are given a collection called Blogs that contains information about blog posts on your website. The fields in each of the documents in this collection are

- (a) `_id` - An ObjectId value
- (b) `'Title'` - Title of the blog post
- (c) `'Category'` - Topic category of the blog post
- (d) `'Likes'` - A numerical value that denotes the number of likes the blog post received

You want to find all the records that belong to the Category Sports and have exactly 200 Likes. Which of the following commands will give you the necessary result?

- `db.Blogs.find('Likes':200,'Category':'Sports')`
- `db.Blogs.find('Category':'Sports','Likes':200)`
- `db.Blogs.find('Category':'Sports')`  
`db.Blogs.find('Likes':200)`
- Both A and B → CORRECT

**Q163.** Which of the following statements is true?

- (a) SQL databases are ideally suited for vertical scaling, whereas NoSQL databases are ideally suited for horizontal scaling.
- (b) The schema in NoSQL databases is flexible.
- (c) NoSQL databases are more suitable than SQL databases for complex querying procedures to get the relevant results.

- 1 and 2  $\rightarrow$  CORRECT
- All of them
- 1 and 3
- 2 and 3

**Q164.** Which entity in MongoDB is similar to tables in SQL?

- Documents
- Database
- Fields
- Collections  $\rightarrow$  CORRECT

**Q165.** Let's say you are currently working in the database **Orders** and want to switch to the database **Sales**. Which MongoDB command do you need to use in this case?

- switch sales  
use sales
- use sales  $\rightarrow$  CORRECT
- switch orders  
use sales
- switch sales  
use sales

**Q166.** You are using the following command in a MongoDB collection:

```
db.products.updateOne(
 { category: "electronics" },
 {
 $inc: { stock: 10 },
 $set: { available: true }
 }
)
```

Which of the following best describes what this command does?

- Increases stock by 10 for all documents and sets available to true
- Sets stock to 10 and available to true for one document
- Increases stock by 10 and sets available to true for the first matching document  $\Rightarrow$  CORRECT
- Updates all documents where category is "electronics"

**Q167.** Which of the following filters will find documents in a books collection where the author field starts with “John” and the price is less than 500 or greater than 1500?

- {author: /^John/, price:{ \$lt:500, \$gt:1500}}
- {author: /^John/, \$or: [{ price:{ \$lt:500}}, {price:{ \$gt:1500}}]}  $\rightarrow$  CORRECT
- { \$and:[{ author: /John/}, {price:{ \$lt:500, \$gt:1500}}]}
- {author:{ \$regex: "^John"}, price:{ \$lte:1500}}

**Q168.** Given an **orders** collection, what will the following aggregation pipeline return?

```
db.orders.aggregate([
 { $match: { status: "shipped" } },
 { $group: { _id: "$customerId", total: { $sum: "$amount" } } }
])
```

- The sum of all orders for each customer, regardless of status
- The sum of order amounts for all customers, grouped as a single total
- The total order amount for each customer where the status is "shipped"  $\rightarrow$  CORRECT
- The average order amount for shipped orders by each customer

**Q169.** In which scenario would placing a `$match` stage after a `$group` stage in an aggregation pipeline be necessary?

Note: Only one option is correct

- When filtering documents based on aggregated values like `total > 1000` → CORRECT
- When filtering documents using fields not present in the `$group` output
- When optimising performance for filtering raw data
- When matching based on original document fields before aggregation